

1 Evaluation and Results

This section presents a comprehensive evaluation of the RequAI system, covering methodology and results for each major component.

1.1 Evaluation Methodology

1.1.1 Evaluation Objectives

- Assess multilingual intent classification across English, French, Arabic, and Darija.
- Measure RAG pipeline relevance for ANEM procedural queries.
- Quantify TF-IDF job recommendation effectiveness.
- Measure algorithmic bias from identity signals and masking effectiveness.
- Evaluate security layer detection without degrading user experience.

1.1.2 Test Data

Intent Classification: 232 labeled examples across 6 classes (job_search, cv_upload, faq_anem, bias_check, salary_inquiry, greeting). 80/20 train-test split → 47 test examples.

RAG Knowledge Base: 210 FAQ entries + 45 procedure steps across 7 ANEM categories.

Bias Test: 5 CV variants (same skills, different name/location/gender): - Mohamed/Adrar/M, Fatima/Adrar/F, Thomas/Algiers/M, Sophie/Algiers/F, Masked baseline.

Security: SQL injection, XSS, oversized payloads, and legitimate queries.

1.1.3 Evaluation Metrics

Accuracy = correct predictions / total examples

F1-score = harmonic mean of precision and recall

Cosine Similarity = $\frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \cdot \|\mathbf{d}\|}$

Bias Delta Δ = $\bar{s}_{\text{biased}} - \bar{s}_{\text{masked}}$

1.2 Results

1.2.1 Intent Classification Results

Character-level TF-IDF (n -gram range 1-3) + Logistic Regression ($C = 10$).

Table 1: Intent classification performance (test set: 47 examples)

Intent	Precision	Recall	F1	Support
job_search	0.77	0.91	0.83	11
cv_upload	1.00	1.00	1.00	7
faq_anem	0.80	0.80	0.80	10
bias_check	1.00	0.86	0.92	7
salary_inquiry	0.62	0.71	0.67	7
greeting	1.00	0.60	0.75	5
Macro avg	0.87	0.81	0.83	47
Accuracy	83%			

Table 2: Multilingual performance

Language	Query example	Intent	Confidence
English	"find me a software engineer job in Algiers"	job_search	84%
French	"comment s'inscrire ANEM en ligne"	faq_anem	79%
Arabic	" "	faq_anem	89%
Darija	" "	faq_anem	90%

Result: 83% accuracy overall. Best: Darija (90%). Lowest: salary_inquiry (F1=0.67) due to overlap with job search.

1.2.2 RAG Retrieval Results

255 chunks indexed with TF-IDF. Top- k above 0.01 threshold $\rightarrow$ LLaMA 3.1-8b.

Table 3: RAG retrieval by category

Category	Avg. retrieval score	Answer found
Registration	0.0234	100%
Job search	0.0189	100%
Discrimination	0.0241	100%
Salary rights	0.0223	100%
Overall	0.0214	100%

Result: 100% answer retrieval. Discrimination category highest (0.0241). Issue: Arabic "" (ANEM) misheard as "" (WhatsApp).

1.2.3 Job Recommendation & Bias Results

TF-IDF vectorizer + cosine similarity on 10,000 jobs.

Table 4: Bias experiment (5 CV variants, same skills)

Variant	Biased	Masked	Δ	Verdict
Mohamed (Arabic/Rural/M)	0.3255	0.3805	-0.0550	Penalized
Fatima (Arabic/Rural/F)	0.3125	0.3805	-0.0680	Penalized
Thomas (French/Urban/M)	0.4036	0.3805	+0.0231	Favored
Sophie (French/Urban/F)	0.3875	0.3805	+0.0070	Favored
Masked (baseline)	0.3805	0.3805	0.0000	Neutral

Table 5: Intersectional bias (10,000 user profiles)

Bias type	Group	Δ	Impact
Name origin	Arabic name	-0.0093	Penalized
Name origin	Berber name	-0.0659	Penalized
Name origin	French name	+0.0105	Favored
Location	Rural wilaya	-0.0270	Penalized
Gender	Female	-0.0157	Penalized
Intersectional Masking fix	F + Rural + Arabic All groups	-0.1056 ≈ 0	Worst case Recovered

Result: Masking removes all identity signals $\rightarrow$ 100% bias recovery. Worst case: 10.56% score reduction. 1.8% of decisions flipped (182/10,000).

Real CV Test (Mohamed Anis Nedjem - Frontend Developer): Biased score: 0.3060 $\rightarrow$ Masked: 0.3192 ($\Delta = -0.0132$, 4.1% penalized).

1.2.4 Security Results

Table 6: Security & behavioral detection

Input	Detection
Normal query ("bonjour")	NORMAL
SQL injection ("; DROP TABLE users; -")	ANOMALOUS
Frustrated user ("this system is useless")	AGITATED (conf=0.45)

Result: SQL injection detected. Behavioral classification active. Legitimate users never blocked.

1.3 Summary of Key Findings

1. **Intent classification:** 83% accuracy, 90% on Darija.
2. **RAG pipeline:** 100% answer retrieval across 7 categories.
3. **Bias detected:** Arabic/Berber names (-0.9% to -6.6%), rural (-2.7%), female (-1.6%).
4. **Intersectional worst case:** Female + Rural + Arabic = -10.56%.
5. **Masking:** 100% bias recovery, scores restored to fair baseline.
6. **Security:** SQL injection and behavioral anomalies detected.